Section A (60 marks)

1 **(a)** Distinguish between *gravitational potential energy* and *elastic potential energy*.

……

……[1]

(b) A child's pogo stick comprises of a spring with spring constant of 1.50×10^4 N m^{-1}. The combined mass of the child and pogo stick is 25.0 kg.

Fig. 1.1 shows the various instances of a child jumping vertically on the pogo stick.

At instance **A**, the spring compression is a maximum and the child is momentarily at rest. The foot rest is 0.200 m below the reference level.

At instance **B**, the spring is relaxed and the child is moving upward.

At instance **C**, the child is again momentarily at rest and at the top of the jump.

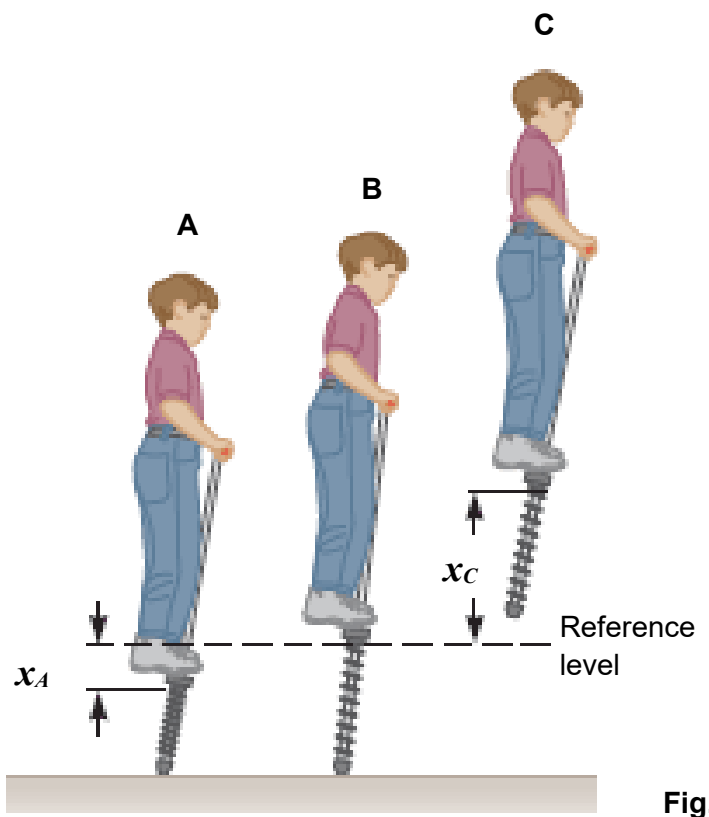

Fig. 1.1

(i) State the energy changes from instance A to C. Numerical values are not required.

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……[3]

[Turn over

(b) **(ii)** Determine the value for x_C.

x_C. =……m [2]

(iii) Determine the maximum speed of the child.

maximum speed = …….. m s^{-1} [3]

(iv) State and explain the position of the footrest, with respect to the reference level, when maximum speed is attained.

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……[1]